

Functional Analysis and Requirements

Westbrook Community Library – Reservation Management System

Version: 1.0

Date: 30 April 2026

Status: Draft – Ready for Stakeholder Review

1. Document Control

1.1 Document Information

Item	Description
Document Title	Functional Analysis and Requirements – Reservation Management System
Project Name	Westbrook Community Library Reservation Management
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1.2 Version History

Version	Date	Author	Description
0.1	20 April 2026	BA Team	Initial structure
0.9	28 April 2026	BA Team	Full draft content
1.0	30 April 2026	BA Team	Consolidated version for review

Executive Summary

Westbrook Community Library serves approximately 45,000 residents across one central building and two branch locations. Currently, all reservations for rooms and shared equipment are managed manually using paper logbooks, telephone calls, and informal emails. This fragmented approach results in

double-bookings, lost records, inconsistent rule enforcement, staff overload, patron dissatisfaction, and limited management visibility.

The objective of this initiative is to define and implement a centralized Reservation Management System that standardizes reservation processes across all three locations. The system must reduce errors, improve operational efficiency, enhance the patron experience, and provide reliable reporting for management decision-making.

The solution must be usable by non-technical staff, accessible from existing branch computers, operate within a limited municipal budget, support at least three simultaneous locations, and avoid prolonged downtime or complex historical data migration.

This document defines the functional and non-functional requirements for the system and serves as the authoritative baseline for design, development, and validation.

2. Introduction

2.1 Purpose

This document defines the complete set of functional and non-functional requirements for the Reservation Management System. It establishes a clear, structured, and traceable baseline to guide development and testing activities.

2.2 Scope

The system will manage reservations for:

- Study rooms
- Meeting halls
- Projectors
- Laptops
- Shared equipment

This document defines **what the system must do**, without prescribing implementation details.

2.3 Intended Audience

- Library Director
- Branch Managers
- Library Staff

- Project Manager
 - Development Team
 - Quality Assurance Team
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3. Organizational Context

Westbrook Community Library operates:

- 1 Central Library
- 2 Branch Locations
- 12 Staff Members

All reservation activities are currently managed manually and independently at each branch. This lack of centralization leads to operational inefficiencies, inconsistent policies, and limited reporting capabilities.

The strategic objective is to introduce a centralized reservation management capability that standardizes processes and improves service quality while respecting budget constraints.

4. Stakeholder Identification

Stakeholder	Role	Interest
Library Director	Executive oversight	Reporting, policy enforcement
Branch Managers	Operational supervision	Visibility and control
Library Staff	Daily operations	Ease of use, reduced workload
Patrons	Service users	Reliable reservations
Municipal Administration	Funding authority	Cost control and accountability

5. Current Situation – AS-IS

Reservations are recorded manually using:

- Paper logbooks
- Telephone communication

- Informal email exchanges

Each branch maintains independent records.

Identified Issues

- Double-bookings due to manual checks
- Lost or illegible records
- No cross-location visibility
- High administrative workload
- Inconsistent reservation rules
- No reliable reporting mechanism

These issues reduce efficiency, increase errors, and limit management insight.

6. TO-BE Vision

The future system will:

- Centralize reservation records across three locations
- Automatically prevent scheduling conflicts
- Standardize reservation rules
- Provide structured reporting
- Improve staff efficiency
- Enhance patron experience

The system will serve as the single authoritative source of reservation data.

7. System Scope and Boundaries

7.1 In Scope

- Management of reservable rooms and equipment
- Creation, modification, and cancellation of reservations
- Multi-location support (minimum three locations)
- Role-based access control
- Reservation reporting
- Audit tracking

7.2 Out of Scope

- Financial transactions or billing
 - Replacement of unrelated library systems
 - Integration with municipal systems
 - Major infrastructure investment
 - Complex historical data migration
 - Expansion beyond current three locations
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8. Functional Requirements

All functional requirements are uniquely identified and traceable.

8.1 User Management

FR-001 – Create User Account

The system shall allow authorized personnel to create staff user accounts.

Priority: High

Source: Management

FR-002 – Assign User Role

The system shall allow assignment of predefined roles to control access permissions.

Priority: High

Source: Management

FR-003 – Authenticate User

The system shall require authentication before access to reservation functions.

Priority: High

Source: Management

FR-004 – Maintain User Account

The system shall allow authorized users to update or deactivate staff accounts.

Priority: Medium

Source: Management

8.2 Resource Management

FR-005 – Register Resource

The system shall allow registration of reservable rooms and equipment.

Priority: High

Source: Staff

FR-006 – Categorize Resource

The system shall allow classification of resources by type.

Priority: High

Source: Staff

FR-007 – Assign Resource to Location

Each resource shall be associated with a specific library location.

Priority: High

Source: Management

FR-008 – Define Resource Availability

The system shall allow definition of standard availability schedules for each resource.

Priority: High

Source: Staff

FR-009 – Update Resource Status

The system shall allow temporary marking of a resource as unavailable.

Priority: Medium

Source: Staff

8.3 Reservation Management

FR-010 – Create Reservation

The system shall allow creation of reservations for available resources.

Priority: High

Source: Staff, Patron

FR-011 – Prevent Double-Booking

The system shall prevent overlapping reservations for the same resource.

Priority: High

Source: Staff

FR-012 – Modify Reservation

The system shall allow modification of existing reservations subject to rules.

Priority: High

Source: Staff

FR-013 – Cancel Reservation

The system shall allow cancellation of existing reservations.

Priority: High

Source: Staff

FR-014 – Search Reservations

The system shall allow searching reservations by date, location, resource, or patron.

Priority: High

Source: Staff

FR-015 – Maintain Reservation History

The system shall maintain a traceable history of reservation actions.

Priority: Medium

Source: Management

FR-016 – View Multi-Location Schedule

The system shall display reservation schedules across all locations.

Priority: High

Source: Management

FR-017 – Enforce Reservation Rules

The system shall enforce defined booking policies such as duration limits.

Priority: High

Source: Management

FR-018 – Capture Patron Information

The system shall require capture of essential patron information during reservation creation.

Priority: High

Source: Staff

8.4 Availability and Scheduling

FR-019 – View Availability by Date

The system shall display availability by selected date.

Priority: High

Source: Staff

FR-020 – Filter Availability by Location

The system shall allow filtering availability by library location.

Priority: High

Source: Staff

FR-021 – Filter Availability by Resource Type

The system shall allow filtering availability by resource category.

Priority: Medium

Source: Staff

8.5 Notifications and Confirmation Records

FR-022 – Generate Reservation Confirmation Record

The system shall generate a standardized confirmation record for each

reservation.

Priority: High

Source: Patron

FR-023 – Record Cancellation Confirmation

The system shall record confirmation of reservation cancellations.

Priority: Medium

Source: Staff

FR-024 – View Upcoming Reservations

The system shall allow viewing of upcoming reservations.

Priority: Low

Source: Staff

8.6 Reporting and Administration

FR-025 – Report Usage by Resource

The system shall generate usage reports by resource.

Priority: Medium

Source: Management

FR-026 – Report Usage by Location

The system shall generate usage reports by location.

Priority: Medium

Source: Management

FR-027 – Report Reservation Volume

The system shall generate reports by time period.

Priority: Medium

Source: Management

FR-028 – Provide Audit Trail Access

The system shall allow authorized users to view logged reservation activities.

Priority: Medium

Source: Management

FR-029 – Configure Reservation Rules

The system shall allow administrators to define and update reservation policies.

Priority: High

Source: Management

FR-030 – Enforce Role-Based Access

The system shall restrict functionality according to assigned user roles.

Priority: High

Source: Management

FR-031 – Retain Reservation Records

The system shall retain reservation records in accordance with defined retention policies.

Priority: Medium

Source: Management

9. Non-Functional Requirements

9.1 Usability

NFR-001 – Ease of Use

Category: Usability

The system shall be intuitive for non-technical staff.

Acceptance: Staff can perform core tasks after basic training.

NFR-002 – Minimal Training

Category: Usability

The system shall require minimal formal training.

Acceptance: Operational readiness within one working day of instruction.

NFR-003 – Clear Error Messages

Category: Usability

The system shall provide understandable error feedback.

Acceptance: Errors clearly describe the issue and corrective action.

9.2 Performance

NFR-004 – Responsive Operations

Category: Performance

The system shall respond promptly to reservation actions.

Acceptance: No noticeable delay during normal use.

NFR-005 – Concurrent Usage

Category: Performance

The system shall support simultaneous use across three locations.

Acceptance: No functional degradation during concurrent operations.

9.3 Availability

NFR-006 – Operational Availability

Category: Availability

The system shall be accessible during library operating hours.

Acceptance: Accessible whenever the library is open.

9.4 Security

NFR-007 – Secure Access Control

Category: Security

Only authorized users shall access reservation functions.

Acceptance: Unauthorized access attempts are denied.

NFR-008 – Patron Data Protection

Category: Security

Patron data shall be protected from unauthorized access.

Acceptance: Only authorized users can view or modify patron data.

9.5 Operational Constraints

NFR-009 – Existing Computer Compatibility

Category: Operational Constraint

The system shall operate on existing branch computers.

Acceptance: No hardware replacement required.

NFR-010 – Limited Budget Compliance

Category: Operational Constraint

The solution shall align with limited municipal budget conditions.

Acceptance: No large capital investment required.

NFR-011 – Minimal Downtime During Transition

Category: Operational Constraint

Implementation shall not cause prolonged service interruption.

Acceptance: Reservation operations remain available during transition.

NFR-012 – No Complex Data Migration

Category: Operational Constraint

The system shall not require full digitization of historical records.

Acceptance: System operational without migrating all past logbooks.

10. Use Cases

(Condensed listing; each includes Actors, Preconditions, Main Flow, Alternative Flow, Postconditions.)

- UC-001 Authenticate User
- UC-002 Check Availability
- UC-003 Create Reservation

- UC-004 Modify Reservation
- UC-005 Cancel Reservation
- UC-006 Manage Resources
- UC-007 Generate Reports
- UC-008 Configure Reservation Rules

Each use case fully aligns with corresponding functional requirements.

11. Assumptions and Dependencies

Assumptions:

- Staff participation in validation
- Standardized policies defined
- Basic computer literacy

Dependencies:

- Policy approval
 - Stakeholder sign-off
 - Data retention decision
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12. Exclusions

- No payment processing
 - No municipal integration
 - No infrastructure redesign
 - No expansion beyond three locations
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13. Open Questions

- Will patrons self-book?
- What are exact booking limits?
- What is the cancellation policy?
- What patron data is mandatory?
- What is the retention period?

14. Glossary

Term	Definition
Reservation	Confirmed allocation of a resource
Resource	Room or equipment available for booking
Patron	Individual requesting reservation
Location	One of three library sites
Audit Trail	Logged record of system actions
Administrator	User with configuration authority
Staff User	Operational system user